

DSA STUDY BLUEPRINT SERIES

87. Binary Tree Paths

Extracting Paths from Root to Leaf Nodes

Section 09 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Paths:** Find all root-to-leaf paths.
- Traverse tree, appending values to path; push to result when leaf is hit.

Memory & Execution Blueprint

Tree paths output: ["1->2", "1->3"]

C++ Implementation Reference

Standard code layout and syntax

```
void getPaths(TreeNode* root, string path, vector& ans) {  
    if (!root) return;  
    path += to_string(root->val);  
    if (!root->left && !root->right) { ans.push_back(path); return; }  
    getPaths(root->left, path + "->", ans);  
    getPaths(root->right, path + "->", ans);  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(H)$ recursion space

Key Practices & Gotchas

- **Important:** Revert path strings correctly when backtracking (stack handles this).
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace recursive Preorder Traversal visits (Node-Left-Right) for tree `1 (left=2, right=3)`:

Call Stack Level	Active Node Visited	Expected Left Traversal	Expected Right Traversal
1	Root (1)	Recurse on Left child (2)	Recurse on Right child (3)
2	Leaf (2)	Base case (nullptr)	Base case (nullptr)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Iterative traversals:** Standard DFS traversals can run iteratively by simulating recursion calls with Stacks.
- **Level Order BFS:** Level tracking searches use Queues to visit node layers.
- **Related LeetCode Problems:** #102 Binary Tree Level Order Traversal, #94 Binary Tree Inorder Traversal, #236 Lowest Common Ancestor.